

Supporting Information for

Non-White scientists appear on fewer editorial boards, spend more time under review, and receive fewer citations

Fengyuan Liu, Talal Rahwan, Bedoor AlShebli

Joint corresponding authors: Talal Rahwan, Bedoor AlShebli
E-mail: talal.rahwan@nyu.edu; bedoor@nyu.edu

This PDF file includes:

Supporting text
Figs. S1 to S9
Tables S1 to S4
SI References

Supporting Information Text

1. Measuring citational distortion

This Supplementary Note contains two sections. The first one shows that, when studying citation, the content of papers provides additional information that cannot be inferred from the venue and discipline of a paper. The second section provides details on how we applied citational lensing, a method recently proposed by Gomez et al. that accounts for the content of papers when studying citation disparity (1), to measure the racial gap of citation rates.

A. Textual similarity predicts citation. Gomez et al. (1) improves over traditional methods of studying citation disparity (2) by using an original measure that accounts for textual similarity between groups of papers when studying citation disparity. However, some may argue that bibliometric information such as venue and discipline of a paper roughly approximates its content, so there might be no need to consider textual information when studying citation.

To examine whether such is the case, we train two classifiers to predict whether a paper cites another given any pair of papers. Both classifiers are ensembles of fifty decision trees boosted using the AdaBoost algorithm (3). The first classifier predicts whether one paper cites another using the discipline, journal, and year of publication of both papers, while the second one considers the content of both papers, and the textual similarity between the two, in addition to the aforementioned bibliometric features. Given a pair of papers, the content of either one of them is represented as a vector of topic distribution obtained using Latent Dirichlet Allocation (LDA), and the textual similarity between the pair is represented as the cosine similarity of those two vectors.

We found that, when the classifier only considers the papers' disciplines, journals, and years of publication, it achieves an accuracy of 68.61% on the test sets. However, if the textual features of a paper is included, the accuracy reaches 85.43%. This suggests that the content of a paper does provide insight that is not fully captured by a paper's venue and discipline, providing further evidence that textual information should be considered when studying citation disparity.

B. Studying citation disparity using citational lensing. Following Gomez et al. (1), we construct multiplex networks consisting of three layers for each discipline t in each year y. The first layer, $\mathbf{L}_{\text{citation}}$, is a network such that each node represents a race, and the weight of an directed edge between node u and v is the number of citations that v receives from all papers that u publishes in discipline t and year y.

The second layer, $\mathbf{L}^{\text{T}}_{\text{text}}$, is a network such that each node represents a race, and the weight of an directed edge between node u and v represents how similar the corpus of papers published by u is similar to that of v in discipline t in year y. Such similarity is calculated as the KL-divergence of the distribution of topics of all papers published by u and by v in the given year and discipline. To obtain the distribution of topics of a each race, we first label each paper by all unique races of authors on that paper, and apply the labelled LDA model (4) to the collection of n-grams of those papers.

Lastly, we standardize the weight of edges within each network, and calculate the third layer as follows: $\mathbf{L}_{\text{distortion}} = \mathbf{L}_{\text{citation}} - \mathbf{L}^{\text{T}}_{\text{text}}$. This layer represents how much more or less one race cites another relative to the textual similarity of research produced by both races.

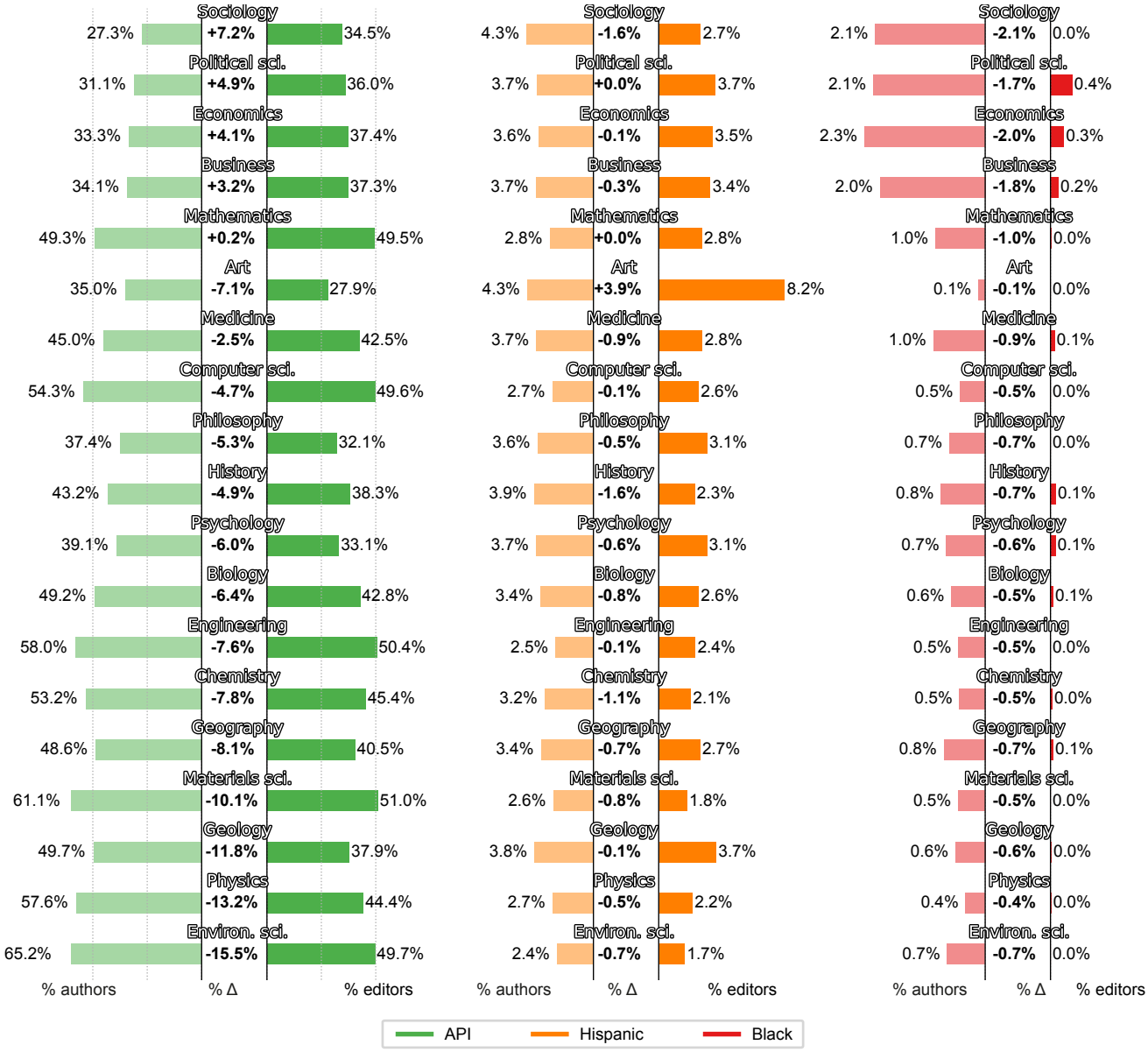


Fig. S1. Representation of races among editors and authors in disciplines. The percentage of API, Hispanic, and Black authorship (left panel) and the percentage of API, Hispanic, and Black editorship (right panel) in each publisher between the years 2011 and 2020 (inclusive). The number at the center represents the difference between the two panels; a positive number indicates that editors of a specific race are overrepresented in the corresponding discipline.

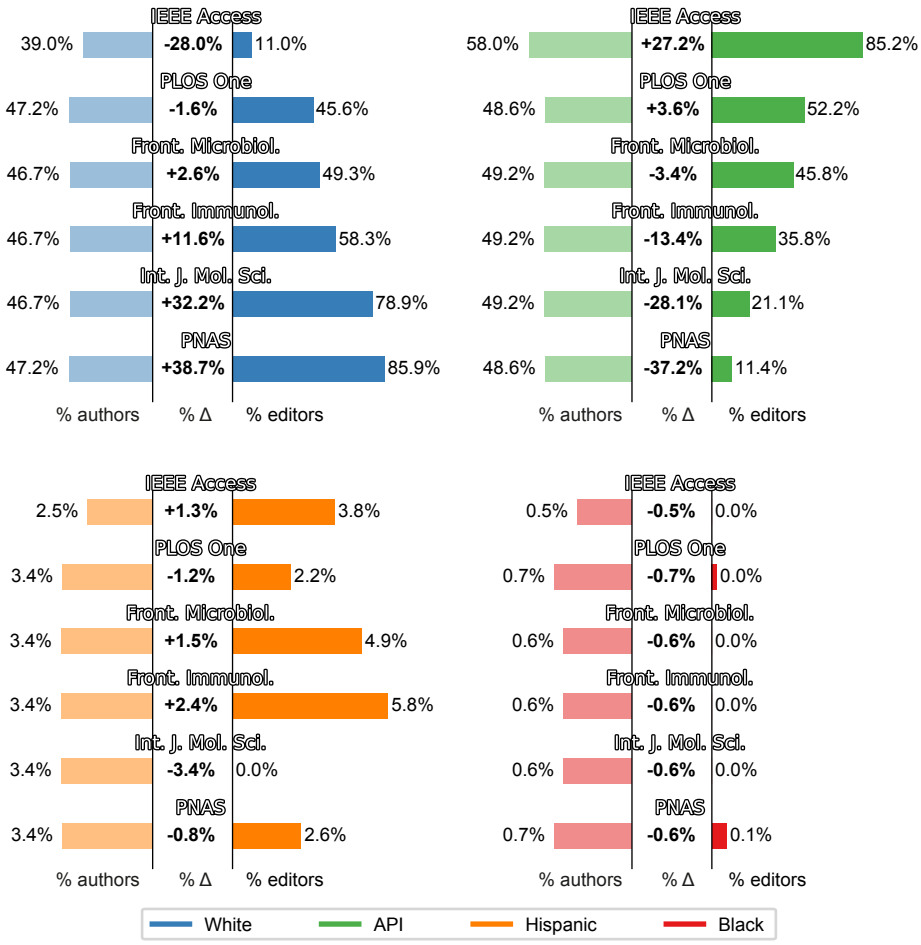


Fig. S2. Representation of races among editors and authors in publishers. The percentage of White, API, Hispanic, and Black authorship (left panel) in each top-ranked journal according to Google Scholar, compared against the percentage of White, API, Hispanic, and Black authors in the journal's discipline (right panel) between the years 2011 and 2020 (inclusive). The number at the center represents the difference between the two panels; a positive number indicates that editors of a specific race are overrepresented in the corresponding journal.

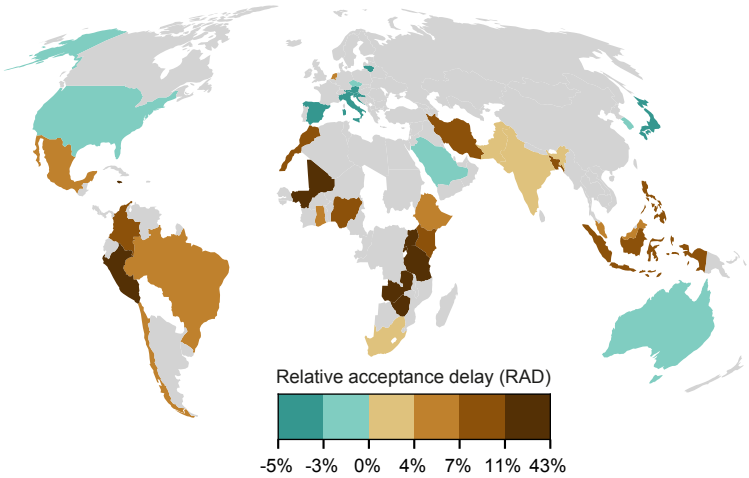


Fig. S3. Average RAD of countries that have at least 10 papers published by the six publishers. The countries whose average RAD is not significantly different from 0 at the 0.01 level are colored in gray. The average RAD of all other countries is divided into 5 quantile intervals, with the interval containing 0 further subdivided into two disjoint intervals, one consisting of values > 0 (assigned a shade of brown) and another consisting of values < 0 (assigned a shade of green).

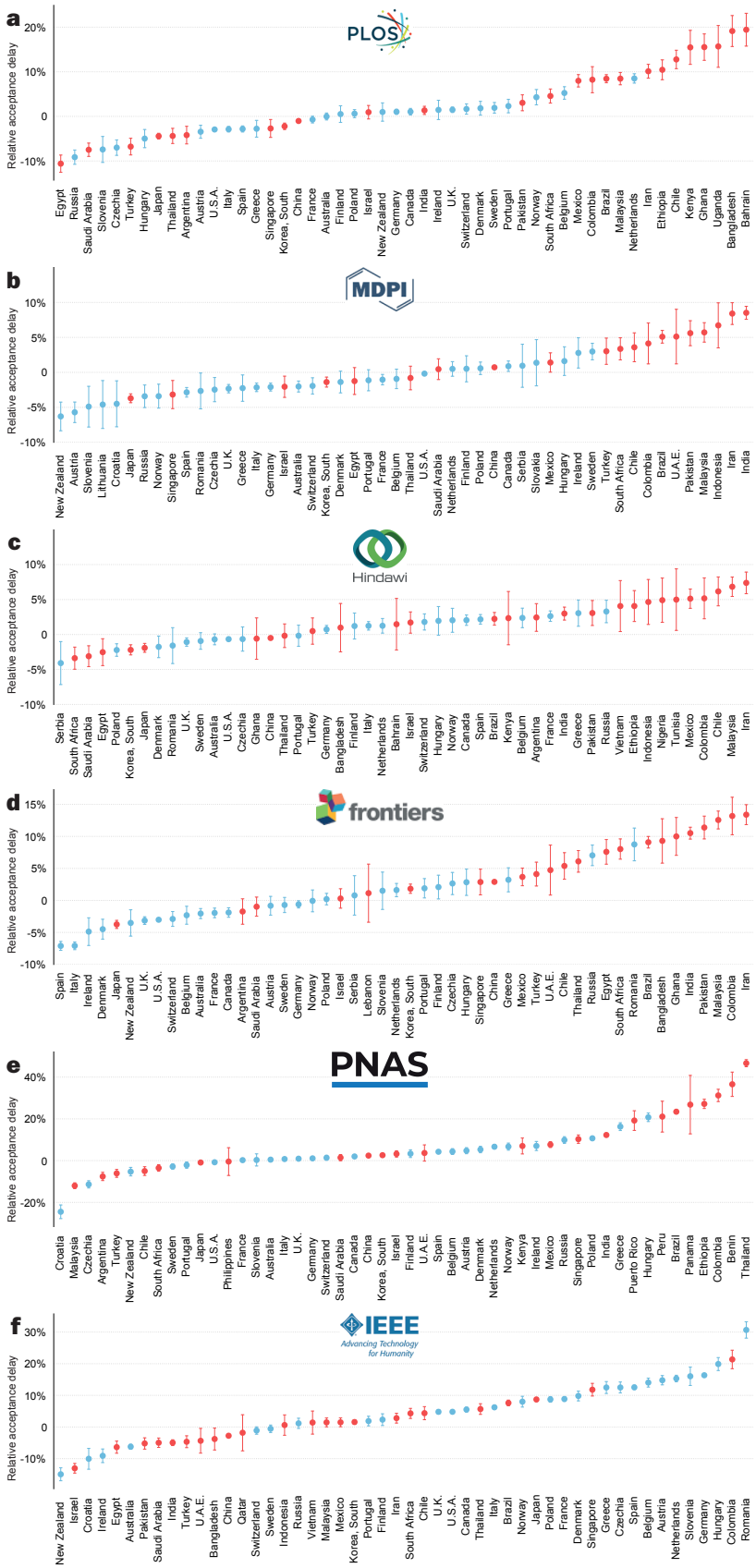


Fig. S4. The average RAD of the 50 countries that with the largest number of papers in each publisher. Results are shown for PLOS (a), MDPI (b), Hindawi (c), Frontiers (d), PNAS (e), and IEEE (f).

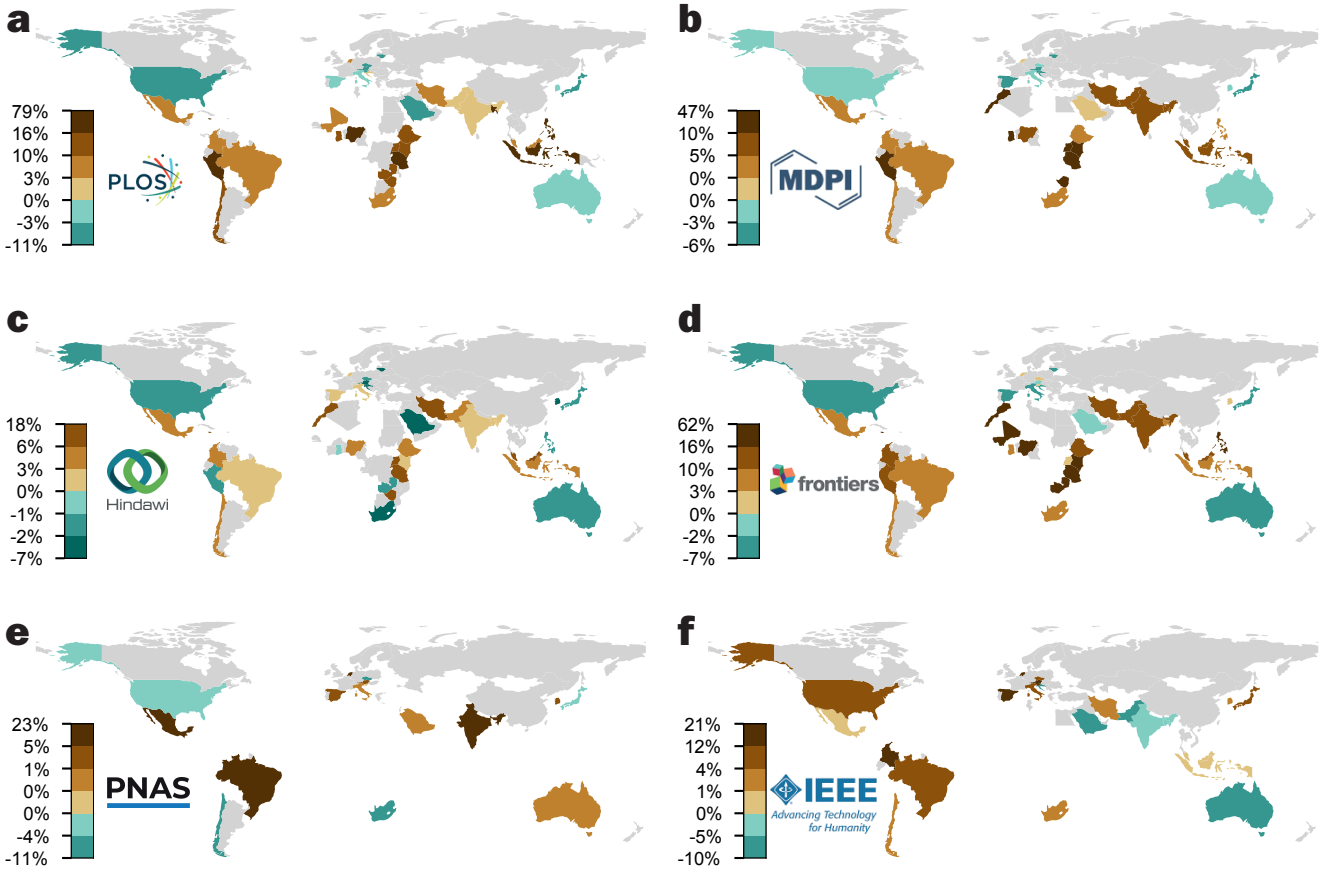


Fig. S5. Countries for which the average RAD is significantly different from 0. Results are shown for PLOS (a), MDPI (b), Hindawi (c), Frontiers (d), PNAS (e), and IEEE (f).

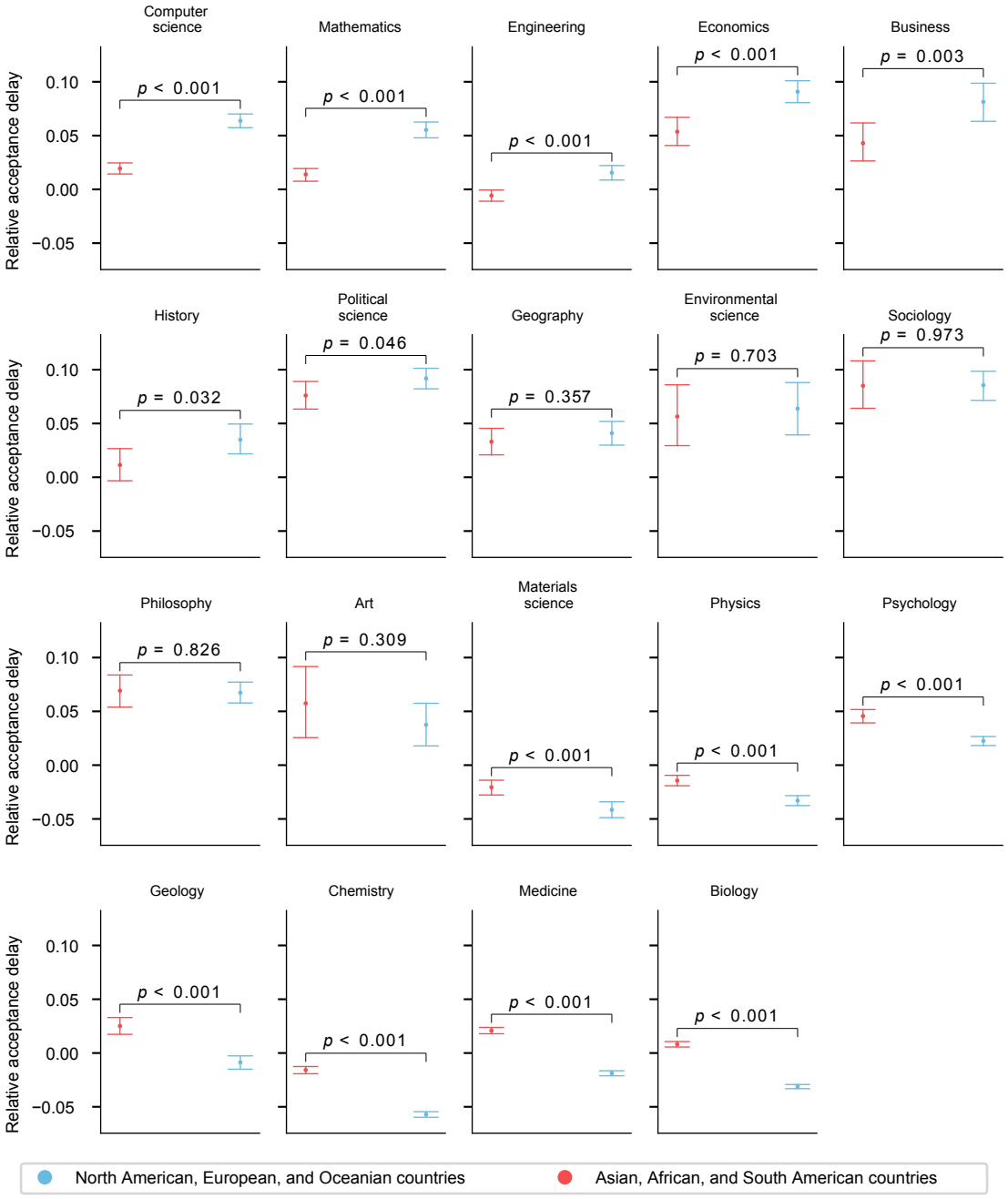


Fig. S6. The relative acceptance delay (RAD) of countries in Asia, Africa, and South America (red), and of those in North America, Europe, and Oceania (blue) in each discipline. Data is presented as mean values ±95% confidence intervals. p-values are calculated using two-sided Welch's t-test.

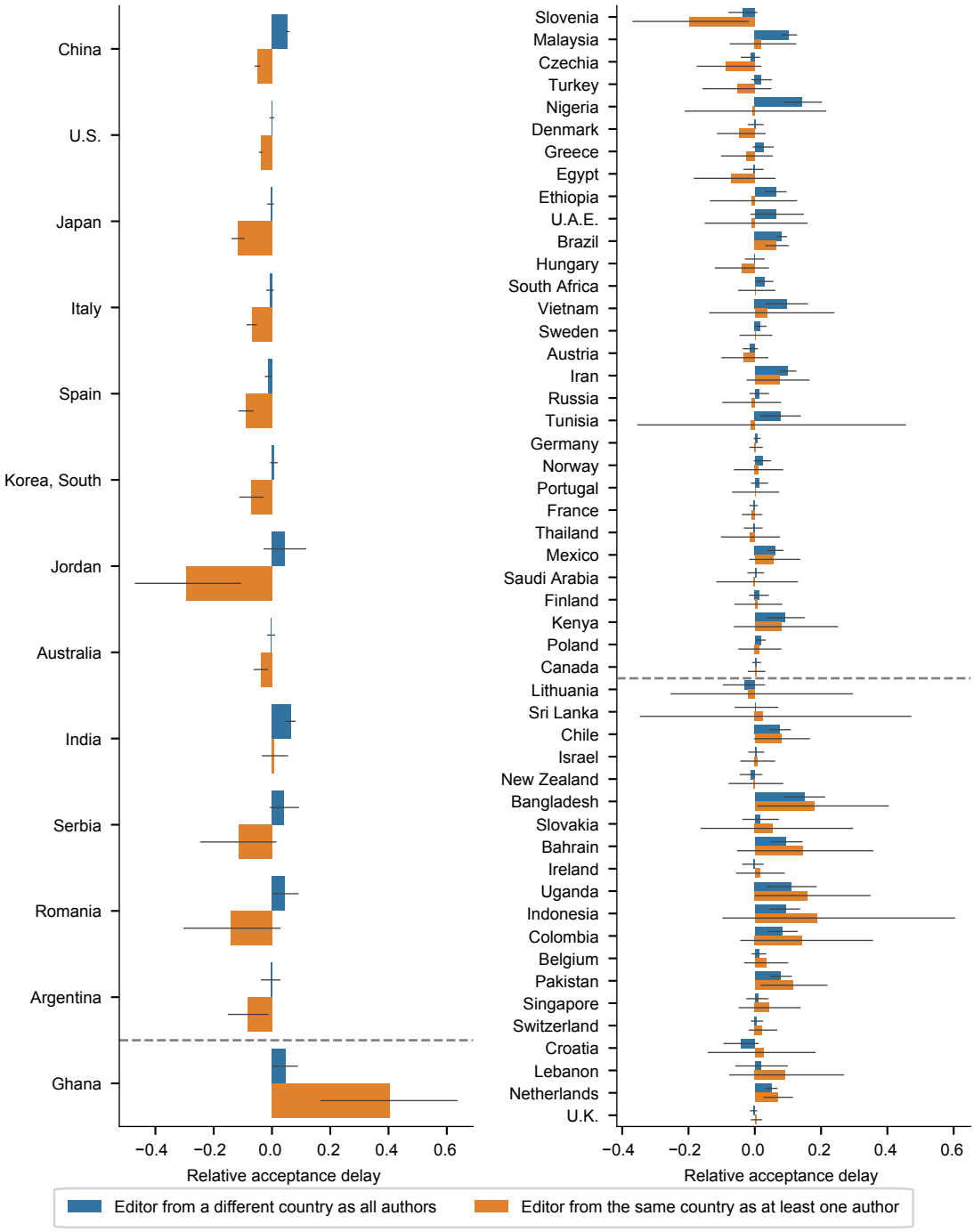


Fig. S7. The relative acceptance delay (RAD) of papers handled by editors from the same country as the authors (orange) and from a different country (blue). Countries in the left panels exhibit a statistically significant difference in RAD between the two groups of papers, while countries on the right panel do not. Countries above (below) the dashed lines are those whose papers experience shorter (longer) RAD when handled by an editor from the same country. Data is presented as mean values ±95% confidence intervals.

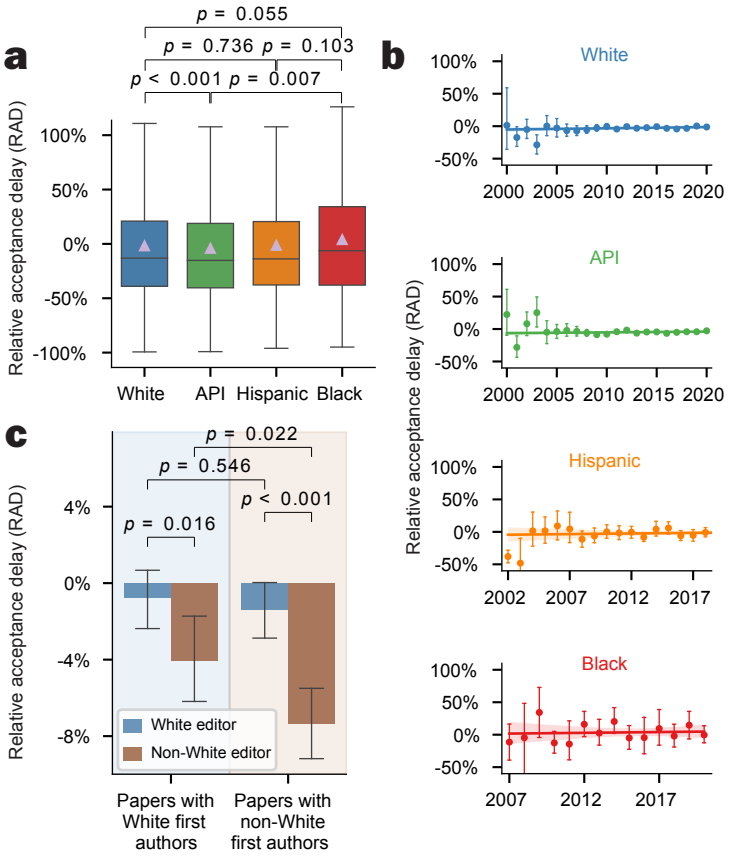


Fig. S8. Racial disparity of relative acceptance delay (RAD). Papers are categorized based on the race of its first author. **a**, RAD distribution of papers with White, API, Hispanic, or Black first authors; mean values are depicted as triangles; boxes extend from the lower to upper quartile values, with a horizontal line at the median; whiskers extend to the most extreme values no further than 1.5 times the interquartile range from the box; p values are calculated using two-sided Welch's t-test. **b**, Average RAD over time for papers with White, API, Hispanic, or Black first authors; lines are fitted using the OLS method, while the shaded region represents 95% confidence intervals of the regression estimate. **c**, RAD of papers handled by editors from the same or different racial group as the first author; p values are calculated using two-sided Welch's t-test.

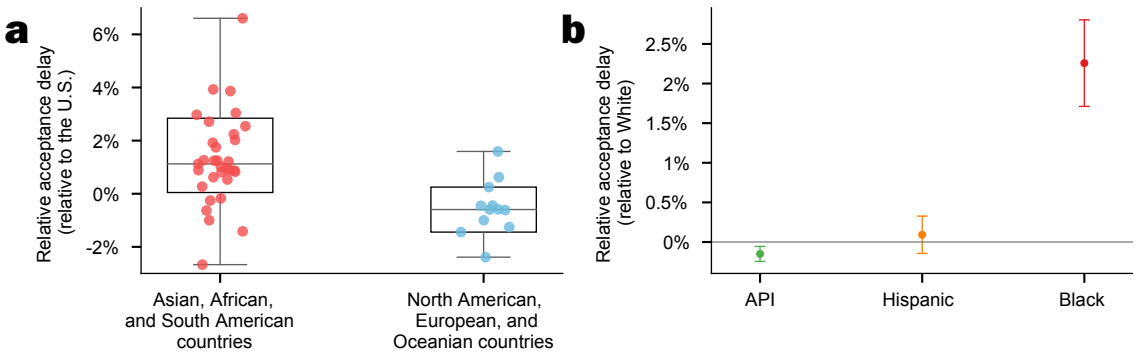


Fig. S9. Regression results of RAD as a function of author and paper characteristics. The linear regression controls for: (i) number of authors from each country, (ii) number of authors from each race, (iii) total number of authors, (iv) maximum academic age of authors, (v) highest affiliation rank of authors, (vi) year of publication, (vii) discipline, (viii) whether the editor is from the same country as the author majority, and (ix) whether the editor is from the same race the author majority. $R^2 = 0.005$, and heteroskedasticity- and autocorrelation-consistent (HAC) standard errors are used. **a**, The distribution of OLS-estimated regression coefficients of country controls that are significant on the 0.05 level. Boxes extend from the lower to upper quartile values, with a horizontal line at the median; whiskers extend to the most extreme values no further than 1.5 times the interquartile range (IQR) from the box. Each data point corresponds to a country. Data outliers outside of the IQR are omitted. **b**, The OLS-estimated regression coefficient of race controls, represented as mean values $\pm$95% confidence intervals.

Table S1. List of papers studying racial and geographical diversity on editorial boards.

Reference	Publication year	Discipline(s) analyzed	No. journals analyzed	Year(s) analyzed
Lim et al. (5)	2022	Cardiology	76	2021
Palser et al. (6)	2022	Psychology & Neuroscience	50	2019
Shim et al. (7)	2021	Psychiatry & Neuroscience	151	2020
Beath et al. (8)	2021	Information System	8	2020
Salazar et al. (9)	2021	Medicine	25	2020
Murray et al. (10)	2019	Biology	1	2012-2017
Ford et al. (11)	2017	Library & Information Science	1	2017
Harzing & Metz (12)	2013	Management	57	1989-2009
Nakayama (13)	2012	Surgery	5	2011
García-Carpintero et al. (14)	2010	Multiple disciplines	281	2005
Braun et al. (15)	2007	Nano Science	16	2005
Polonsky et al. (16)	2006	Marketing	20	1999-2003
Svensson (17)	2005	Marketing	1	2000-2004

Country	% editors	% authors	% editors / % authors
U.S.A.	36.8091%	22.1546%	1.66
China	7.5205%	21.6074%	0.35
Germany	3.7866%	5.1118%	0.74
U.K.	7.1077%	4.7419%	1.5
Italy	7.809%	3.9166%	1.99
Japan	2.5391%	3.6518%	0.7
France	2.7934%	3.2856%	0.85
Spain	3.0464%	2.6853%	1.13
Canada	3.295%	2.5759%	1.28
Australia	4.3109%	2.5642%	1.68
Brazil	1.5506%	2.3425%	0.66
Korea, South	0.6076%	2.2288%	0.27
Netherlands	1.3175%	1.7946%	0.73
India	1.2431%	1.3945%	0.89
Poland	0.6513%	1.2821%	0.51
Switzerland	1.3842%	1.2435%	1.11
Sweden	0.9527%	1.1612%	0.82
Belgium	0.7769%	0.9093%	0.85
Mexico	0.3878%	0.7459%	0.52
Austria	0.84%	0.6982%	1.2
Denmark	0.5138%	0.6962%	0.74
Malaysia	0.2461%	0.6842%	0.36
Israel	0.6146%	0.5933%	1.04
Saudi Arabia	0.3013%	0.5721%	0.53
Norway	0.4141%	0.5658%	0.73
Portugal	0.8865%	0.5657%	1.57
Russia	0.3344%	0.5364%	0.62
South Africa	0.6216%	0.5321%	1.17
Iran	0.3224%	0.4876%	0.66
Finland	0.4031%	0.4863%	0.83
Pakistan	0.2789%	0.4761%	0.59
Czechia	0.2663%	0.4526%	0.59
Singapore	0.4806%	0.4074%	1.18
Thailand	0.0911%	0.4063%	0.22
Chile	0.2441%	0.3684%	0.66
Hungary	0.4806%	0.3645%	1.32
Greece	1.0105%	0.3552%	2.84
Argentina	0.2038%	0.3296%	0.62
Egypt	0.2001%	0.3013%	0.66
New Zealand	0.478%	0.3003%	1.59
Ireland	0.3713%	0.2918%	1.27
Turkey	0.4452%	0.2783%	1.6
Ethiopia	0.0502%	0.2539%	0.2
Romania	0.1669%	0.2137%	0.78
Ghana	0.0212%	0.1643%	0.13
Bangladesh	0.0449%	0.162%	0.28
Colombia	0.0685%	0.1532%	0.45
Serbia	0.1013%	0.1368%	0.74
Kenya	0.0573%	0.1323%	0.43
Indonesia	0.0195%	0.1316%	0.15
Nigeria	0.0408%	0.13%	0.31
Vietnam	0.0431%	0.119%	0.36
Slovenia	0.12%	0.1129%	1.06
Bahrain	0.0437%	0.1096%	0.4
Uganda	0.0294%	0.1079%	0.27
U.A.E.	0.121%	0.1007%	1.2
Croatia	0.0505%	0.0985%	0.51
Slovakia	0.0484%	0.0953%	0.51

Lithuania	0.0298%	0.0776%	0.38
Estonia	0.0952%	0.0746%	1.28
Tunisia	0.005%	0.0712%	0.07
Lebanon	0.0666%	0.0658%	1.01
Cameroon	0.0181%	0.0657%	0.28
Puerto Rico	0.0427%	0.0636%	0.67
Tanzania	0.0185%	0.0571%	0.32
Jordan	0.0272%	0.0547%	0.5
Benin	0.0067%	0.0542%	0.12
Sri Lanka	0.0049%	0.0535%	0.09
Qatar	0.0602%	0.0529%	1.14
Philippines	0.0312%	0.049%	0.64
Peru	0.0809%	0.0488%	1.66
Bulgaria	0.0603%	0.0455%	1.33
Nepal	0.0151%	0.0379%	0.4
Morocco	0.0018%	0.0372%	0.05
Ukraine	0.0195%	0.0311%	0.63
Cyprus	0.0685%	0.0284%	2.41
Ecuador	0.0093%	0.0283%	0.33
Iraq	0.0022%	0.0277%	0.08
Kuwait	0.0335%	0.0271%	1.24
Uruguay	0.0206%	0.0253%	0.81
Iceland	0.029%	0.0242%	1.2
Venezuela	0.0019%	0.0232%	0.08
Zambia	0.0122%	0.023%	0.53
Zimbabwe	0.008%	0.022%	0.37
Malawi	0.0089%	0.022%	0.41
Mali	0.0026%	0.0208%	0.12
Latvia	0.0104%	0.0207%	0.5
Luxembourg	0.0088%	0.0206%	0.43
Oman	0.0237%	0.0193%	1.23
Kazakhstan	0.0439%	0.0183%	2.4
Cambodia	0.01%	0.0181%	0.55
Sudan	0.0008%	0.0169%	0.05
Saint Kitts And Nevis	0.0215%	0.0159%	1.35
Jamaica	0.0049%	0.0153%	0.32
Cuba	0.0167%	0.0152%	1.1
Panama	0.0076%	0.0145%	0.53
Antigua And Barbuda	0.0034%	0.0143%	0.24
Senegal	0.0005%	0.011%	0.05
Reunion	0.0003%	0.0103%	0.03
Algeria	0.0026%	0.0096%	0.27
Congo (Kinshasa)	0.0091%	0.0091%	1.0
Malta	0.0168%	0.0085%	1.98
Bosnia And Herzegovina	0.0027%	0.0084%	0.32
Mozambique	0.0019%	0.008%	0.24
Botswana	0.0126%	0.0079%	1.58
Syria	0.0005%	0.0075%	0.07
Papua New Guinea	0.0014%	0.0073%	0.2
Belarus	0.0039%	0.0071%	0.55
Armenia	0.0003%	0.0067%	0.04
Yemen	0.0004%	0.0065%	0.06
Rwanda	0.0025%	0.006%	0.41
Laos	0.001%	0.0058%	0.18
Montenegro	0.0038%	0.0052%	0.72
French Guiana	0.0036%	0.0051%	0.71
Grenada	0.0026%	0.0045%	0.58
Mongolia	0.0023%	0.0044%	0.53
Brunei	0.0023%	0.0043%	0.54
Georgia	0.0035%	0.0041%	0.85
Libya	0.0012%	0.004%	0.29

Mauritius	0.0049%	0.0035%	1.39
Macedonia	0.0001%	0.0035%	0.04
Guatemala	0.0004%	0.0033%	0.12
Gabon	0.0004%	0.003%	0.13
Madagascar	0.0006%	0.0028%	0.23
Uzbekistan	0.0003%	0.0026%	0.1
Azerbaijan	0.0026%	0.0026%	0.99
Niger	0.0023%	0.0026%	0.9
Costa Rica	0.0001%	0.0026%	0.05
Albania	0.0012%	0.0024%	0.49
Fiji	0.0014%	0.0023%	0.62
Burma	0.0017%	0.0023%	0.73
Guinea-Bissau	0.0006%	0.0019%	0.34
Dominican Republic	0.0009%	0.0016%	0.57
Paraguay	0.0012%	0.0016%	0.75
Honduras	0.0004%	0.0013%	0.3
Swaziland	0.0001%	0.0012%	0.11
Nicaragua	0.0018%	0.0011%	1.72
Afghanistan	0.0001%	0.0009%	0.14
Guam	0.001%	0.0008%	1.25
Moldova	0.0021%	0.0007%	2.95
French Polynesia	0.0003%	0.0006%	0.45
Angola	0.0017%	0.0005%	3.54
Liberia	0.0006%	0.0003%	2.09
Guyana	0.0004%	0.0002%	1.56
El Salvador	0.0003%	0.0002%	1.25

Table S2. The share of countries among authorships and editorships. The table only includes countries that are represented both among the authors and on the editorial boards. Countries are sorted based on the percentage of authorships.

Country	Average RAD	No. papers
U.S.A.	-2.12%	170104
China	0.3%	168039
Germany	0.02%	38358
U.K.	-0.32%	34582
Italy	-3.02%	27314
Japan	-3.37%	26687
Canada	0.49%	21950
Spain	-2.89%	21518
France	-0.78%	20077
Korea, South	-1.15%	19623
Australia	-1.05%	19503
Brazil	6.81%	15879
India	4.23%	13428
Netherlands	5.06%	13047
Poland	0.16%	11934
Sweden	0.93%	9028
Switzerland	-0.26%	8848
Belgium	1.57%	6476
Malaysia	7.1%	6411
Mexico	4.44%	5761
Israel	0.48%	5192
Iran	8.5%	5128
Austria	-2.76%	5002
Saudi Arabia	-2.53%	4996
Denmark	-0.41%	4951
South Africa	3.18%	4643
Portugal	0.58%	4559
Norway	1.91%	4105
Russia	-0.82%	3956
Pakistan	3.36%	3926
Thailand	-0.73%	3548
Finland	1.14%	3421
Czechia	-2.31%	3341
Turkey	0.11%	3195
Chile	6.69%	2910
Greece	0.35%	2860
Egypt	-2.45%	2851
New Zealand	-1.78%	2542
Hungary	0.55%	2508
Argentina	-2.02%	2492
Ethiopia	7.29%	2487
Ireland	-0.88%	2316
Romania	-0.12%	1777
Colombia	7.71%	1316
Ghana	6.07%	1296
Nigeria	9.16%	1259
Bangladesh	9.12%	1146
Slovenia	-3.86%	1097
Indonesia	8.59%	1094
Serbia	0.12%	1057
Vietnam	3.65%	968
Kenya	10.75%	901
U.A.E.	1.62%	875
Croatia	-4.52%	765
Slovakia	-0.21%	753
Lithuania	-4.6%	689
Tunisia	3.93%	588
Uganda	12.25%	581
Jordan	-1.21%	540

Lebanon	-0.26%	515
Sri Lanka	1.78%	515
Puerto Rico	3.17%	489
Estonia	0.05%	485
Cameroon	5.22%	459
Bulgaria	-6.12%	390
Tanzania	16.63%	383
Qatar	3.37%	379
Morocco	8.4%	339
Philippines	8.79%	319
Nepal	4.6%	318
Iraq	-1.17%	305
Kuwait	-3.04%	300
Ukraine	1.7%	285
Benin	0.7%	282
Peru	15.03%	275
Cyprus	0.15%	231
Ecuador	3.25%	219
Uruguay	-4.54%	184
Zimbabwe	10.95%	180
Latvia	-0.38%	176
Oman	-0.21%	157
Luxembourg	7.59%	152
Venezuela	7.12%	143
Kazakhstan	-7.51%	143
Iceland	4.68%	141
Jamaica	15.08%	132
Zambia	12.08%	122
Malawi	16.6%	115
Sudan	3.95%	112
Cambodia	6.79%	106
Mali	15.88%	99
Cuba	8.99%	86
Algeria	4.61%	82
Panama	10.75%	78
Syria	8.43%	77
Bosnia And Herzegovina	0.86%	65
Botswana	7.36%	64
Senegal	3.03%	58
Yemen	8.88%	55
Armenia	-0.63%	51
Belarus	3.81%	50
Montenegro	-4.55%	45
Congo (Kinshasa)	13.23%	42
Rwanda	42.55%	38
Mozambique	29.18%	38
Georgia	2.92%	34
Macedonia	3.2%	32
Brunei	1.79%	32
Azerbaijan	11.44%	31
Fiji	20.27%	25
Papua New Guinea	22.45%	25
Togo	0.7%	24
Côte D'Ivoire	-1.65%	21
Uzbekistan	-0.28%	20
Libya	0.59%	19
Burkina Faso	-2.09%	19
Laos	-9.47%	19
Mongolia	23.61%	18
Swaziland	18.8%	16
Costa Rica	22.98%	16

Guatemala	7.38%	14
Burma	10.94%	14
Madagascar	15.94%	13
Congo (Brazzaville)	-16.1%	13
Niger	-9.36%	12
Korea, North	19.74%	11
Albania	-8.81%	10
Guinea-Bissau	32.89%	10

Table S3. The average RAD of countries around the world. The table only includes countries where at least ten papers have an majority of authors with an affiliation in that country. Countries are sorted based on the number of papers.

Table S4. Number of papers, editorships, and authorships in each publisher.

Publisher	No. papers	No. authorships	No. editorships
PLOS	285502	1981433	177411
MDPI	249047	1310646	96729
Hindawi	208678	959212	97602
Frontiers	204582	1242286	138727
PNAS	45837	330699	37916
IEEE	40781	172022	12927

References

1. CJ Gomez, AC Herman, P Parigi, Leading countries in global science increasingly receive more citations than other countries doing similar research. *Nat. Hum. Behav.* pp. 1–11 (2022).
2. MW Nielsen, JP Andersen, Global citation inequality is on the rise. *Proc. Natl. Acad. Sci.* **118**, e2012208118 (2021).
3. Y Freund, RE Schapire, A decision-theoretic generalization of on-line learning and an application to boosting. *J. computer system sciences* **55**, 119–139 (1997).
4. D Ramage, D Hall, R Nallapati, CD Manning, Labeled lda: A supervised topic model for credit attribution in multi-labeled corpora in *Proceedings of the 2009 conference on empirical methods in natural language processing*. pp. 248–256 (2009).
5. OZ Lim, et al., Disparity in female and asian representation amongst cardiology journal editorial boards members: a call for empowerment. *QJM: An Int. J. Medicine* (2022).
6. ER Palser, M Lazerwitz, A Fotopoulou, Gender and geographical disparity in editorial boards of journals in psychology and neuroscience. *Nat. Neurosci.* **25**, 272–279 (2022).
7. RS Shim, LM Tully, G Yu, EC Monterozza, M Blendermann, Race and ethnicity of editorial board members and editors as an indicator of structural racism in psychiatry and neuroscience journals. *JAMA psychiatry* **78**, 1161–1163 (2021).
8. C Beath, Y Chan, RM Davison, AR Dennis, JC Recker, Editorial board diversity at the basket of eight journals: A report to the college of senior scholars. *Commun. Assoc. for Inf. Syst. CAIS* **48** (2021).
9. JW Salazar, JD Claytor, AR Habib, V Guduguntla, RF Redberg, Gender, race, ethnicity, and sexual orientation of editors at leading medical and scientific journals: a cross-sectional survey. *JAMA Intern. Medicine* **181**, 1248–1251 (2021).
10. D Murray, et al., Author-reviewer homophily in peer review. *BioRxiv* p. 400515 (2019).
11. E Ford, WA Kaspar, P Seiden, Diversity of ACRL publications, editorial board demographics: A report from ACRL's Publications Coordinating Committee. *Coll. & Res. Libr. News* **78**, 548 (2017).
12. AW Harzing, I Metz, Practicing what we preach. *Manag. Int. Rev.* **53**, 169–187 (2013).
13. DK Nakayama, Asian americans in leadership positions in academic surgery. *Annals surgery* **255**, 583–588 (2012).
14. E García-Carpintero, B Granadino, L Plaza, The representation of nationalities on the editorial boards of international journals and the promotion of the scientific output of the same countries. *Scientometrics* **84**, 799–811 (2010).
15. T Braun, S Zsindely, I Dióspatonyi, E Zádor, Gatekeeping patterns in nano-titled journals. *Scientometrics* **70**, 651–667 (2007).
16. MJ Polonsky, R Garma, JD Mittelstaedt, An examination of the globalisation of authorship in publishing in 20 leading marketing journals. *Eur. Bus. Rev.* (2006).
17. G Svensson, Ethnocentricity in top marketing journals. *Mark. Intell. & Plan.* (2005).